



Introduction to Engineering Design Challenge

College - Introductory Engineering • Engineering / CTE
Project Packet

Original Practice • Assessment • Complete Teacher Key



Teacher Guide

Level: College - Introductory Engineering

State: Not applicable

Subject: Engineering / CTE

Resource type: Project Packet

Estimated time: 60-90 minutes

Standard / objective: Course objective - Define a design problem, distinguish criteria from constraints, generate and compare concepts, plan a fair test, and justify an evidence-based iteration.

Source note: Course-level learning objective; no accreditation, transfer, or universal equivalency claimed.

What is included: branded cover, teacher guide, student directions, 16 original practice items, 5-item exit ticket, complete answer key/worked guidance, differentiation supports, printing notes, and license statement.

Purpose: Design a low-cost paper structure that supports a specified load while documenting decisions and test evidence.

Preparation: Print student pages single-sided or assign through a secure LMS. Review the key and select tools allowed for your learners. No special commercial curriculum is required.

Access and differentiation

AIG: require a second representation, proof, counterexample, transfer task, or student-created extension. **504:** provide extended time, chunked sections, breaks, reduced copying, and accessible digital zoom as specified by the plan. **EC:** preteach vocabulary, model one item, provide manipulatives/visuals, and accept aligned response modes from the IEP. **ESL:** preview key terms, provide bilingual resources and sentence frames, and allow partner rehearsal. Use each learner's current plan; this guide does not replace it.



Student Practice

Learning focus: Design a low-cost paper structure that supports a specified load while documenting decisions and test evidence.

Directions: Complete all items. Show calculations, cite evidence, or explain your reasoning when requested. Check your work before submitting.

1. Define criterion.
2. Define constraint.
3. Given max 10 sheets, identify the constraint.
4. Given support 500 g, identify the criterion.
5. Why generate multiple concepts?
6. Name a fair-test variable to control.
7. What is a prototype?
8. Why record failure location?



Student Practice - Continued

9. Create a weighted decision matrix with three criteria.
10. What makes test evidence reliable?
11. Distinguish accuracy from precision.
12. Suggest one safety rule.
13. How can triangles strengthen structures?
14. State a testable claim about folded paper.
15. Use evidence to justify an iteration.
16. Reflect on tradeoffs.



Exit Ticket

Complete independently in 8-10 minutes.

1. State the main skill or concept in your own words.
2. Create one accurate example that uses today's skill.
3. Explain how you checked your example.
4. Identify one common error and correct it.
5. Rate your confidence from 1-4 and name your next practice step.

Quick rubric: 4 = accurate, complete, and well justified; 3 = accurate with minor omission; 2 = partial understanding; 1 = limited evidence; 0 = blank or unrelated.



Complete Teacher Answer Key

1. A desired feature or measure of success.
2. A limit the design must obey.
3. Use no more than 10 sheets.
4. Support at least 500 g.
5. To compare alternatives before committing.
6. Same load placement, materials, span, or test time.
7. A testable representation of a design idea.
8. It reveals where and how to improve the design.
9. Answers vary; criteria, weights, scores, totals required.
10. Consistent procedure and repeat trials.
11. Accuracy is closeness to true value; precision is repeatability.
12. Keep loads low, protect hands/eyes, and clear the test area.
13. They resist shape change and distribute loads.
14. Answers vary; must identify design change and measurable outcome.
15. Cite test result, failure mode, and planned change.
16. Explain how improving one criterion may reduce another.

Exit ticket guidance: Responses vary. Award full credit when the student accurately explains the focus, supplies a correct example, describes a valid check, identifies and corrects a realistic misconception, and chooses a specific next step.

Verification: Fixed responses and calculations were checked during production. Open responses are evaluated against the criteria above.

Originality and use: This packet contains original educational content. It does not reproduce released-test items or branded curriculum. One purchaser may use it with their own students; redistribution, resale, and public posting are not permitted.